

Ziyuan ZHOU

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Contact Information

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EDUCATION

ENS de Lyon

PhD in Mathematical Physics

ETH Zürich

MSc in Physics with Distinction GPA: 5.85/6.00

Quantum Field Theory II, String Theory, General Relativity, Integrability, Infinite-Dimensional Lie Algebras, Conformal Field Theory, Superconductivity, Phenomenology of Particle Physics II, Quantum Chromodynamics, Theoretical Cosmology, etc.

Wuhan University

BSc in Physics GPA: 3.79/4.00

Classical Mechanics, Electrodynamics, Quantum Mechanics I, Thermodynamics, Atomic Physics and Nuclear Physics, Physics Lab, Astronomy, Linear Algebra, Calculus, Ordinary Differential Equations, Probability and Statistics, etc.

ETH Zürich

Visiting student at Department of Physics

Quantum Field Theory I, Quantum Mechanics II, Statistical Physics, Group Theory, Phenomenology of Particle Physics I, Solid State Theory, Research Projects

Lyon, France

Oct 2026 - present

Zürich, Switzerland

Sept 2023 - Apr 2026

Wuhan, China

Aug 2019 - June 2023

Zürich, Switzerland

Sept 2021 - Sept 2022

RESEARCH & PROJECT EXPERIENCE

Inverse scattering method and infinite length limit of lattice models

Supervisor: Prof. NIKLAS BEISERT

ETH Zürich

Sept 2024 - Apr 2025

Master's thesis: explored the infinite-length limit of the classical spectral curve and its connection to the inverse scattering method for the Heisenberg spin chain and the Toda chain

Proseminar on mathematical physics: *The Amplituhedron*

Supervisors: Dr. JOHANNES BROEDEL, Dr. MARTIJN HIDDING

ETH Zürich

Feb 2024 - May 2024

Guided self-study on the topic *The Amplituhedron* (primarily based on the original paper *JHEP* 10 (2014) 030)

Feasibility study of search for tau-lepton-enriched semi-visible jets

Supervisor: Prof. ANNAPAOLA DE COSA

ETH Zürich & CERN

Jul 2022 - Dec 2022

Contributes to the publication [1]

Sensitivity study of VBF production of semi-visible jets at hadron colliders

Supervisor: PROF. ANNAPAOLA DE COSA

ETH Zürich

Nov 2021 - Feb 2022

Reported preliminary results on the sensitivity study of vector boson fusion production of semi-visible jets at the LHC

Search for new physics through rare decays of charm mesons

Supervisor: Prof. LIANG SUN

Wuhan University

Sept 2020 - Sept 2021 & Sept 2022 - May 2023

Research training: measured the masses of the D^+ and K_S mesons

Bachelor's thesis: measured branching ratios of rare decays, e.g. $D_S^+ \rightarrow \rho^+ \phi (\rightarrow e^+ e^-)$

WORKSHOPS & SCHOOLS

Student Workshop on Integrability

Lectures, talks, and posters on diverse topics in integrability, quantum circuits, and more

EU COST Action, Budapest

Jun 2025

Mathematical Physics Summer School (MAPSS)

Courses on Cohomology, Supergeometry, Symplectic Geometry, Hamiltonian Reduction and more

SwissMAP, Les Diablerets

Jul 2024

Balanced Network Combined with Retinal Theory

Six-week Cambridge Research Programme in Theoretical Neuroscience

University of Cambridge, Online

Aug 2021 - Sept 2021

PUBLICATION LIST

- [1] H. Beauchesne, C. Cazzaniga, A. de Cosa, C. Doglioni, T. Fitschen, G. G. di Cortona, and Z. Z. "Uncovering tau leptons-enriched semi-visible jets at the LHC". In: *Eur. Phys. J. C* 83.7 (2023), p. 599. arXiv: 2212.11523 [hep-ph].

HONORS & AWARDS

Scholarship for Outstanding Student in 2019-2020

Wuhan University, Dec 2020

Outstanding Student Award in 2019-2020 Academic Year

Wuhan University, Dec 2020

LuoJia Eminent Exchange Student Scholarship

Wuhan University, Oct 2022

LANGUAGE SKILL

• **Mandarin:** Native • **English:** C1 (IELTS 8) • **German:** Basic

COMPUTING SKILL

Languages: C/C++, Python, Julia, Wolfram, L^AT_EX, etc.

Tools: CERN ROOT, Matlab, Mathematica, GIT, etc.

VOLUNTEER EXPERIENCE

Youth Volunteer Association

Member of Youth Volunteer Association

Wuhan University

Oct 2019-June 2023

HOBBIES/OTHER

Piano, Badminton, Swimming